

Question 1

Consider country Alpha, where the current and the future expected inflation rates are both zero, so that real and nominal interest rates are equal. In period $t + 1$ ("the future"), consumption depends just on disposable income at $t + 1$, and investment on the interest rate and output at $t + 1$ only. Turning now to the current period t , the present, consumption is increasing in both current and expected future disposable income, $C_t = C(Y_t - T_t, Y_{t+1}^e - T_{t+1}^e)$; investment is increasing in current and future expected income, and decreasing in the current and future expected short-term interest rates; finally, G and T are exogenous as usual. The IMF today (time t) publishes new statistics revising downwards the estimated demand for the future.

Suppose that only one-year bonds and two-year bonds exist and that, before the announcements described above, at time t the yield curve was upward sloping. Moreover, assume that the central bank announces today that the bank will enact any policy that would result to be necessary in the future to keep constant the level of GDP. What will be the effects of the new forecasts and the CB's policy announcement on the position and the slope of the yield curve of the country? Explain your answer.

Question 2

Consider an economy, Macrolandia, freely trading goods, services and financial assets with the rest of the world (Foreign country), in a flexible exchange rate regime. Domestic and foreign prices are constant, and equal to one ($P = P^* = 1$).

Using the pair of graphs 'open economy IS-LM - interest parity condition', suppose now that the foreign country adopts a policy mix such that the foreign interest rate, i^* , increases, while foreign GDP, Y^* , remains constant.

Describe the impact of the policy change in the foreign country on the economy of Macrolandia and provide the economic intuition.

Moreover, assume that before the shock Macrolandia's trade were balanced ($NX = 0$). What happens to net exports after the shock?

Question 3

True or False?

Explain whether the following statements are true or false. Motivate your answer in a brief but rigorous way, by making explicit reference to the relevant theory. Lack of proper explanations will result in zero points.

Consider a country that, in period $t = 1$ inherits from the previous period $t = 0$ a debt-to-GDP ratio of 100%: $b_0 \equiv B_0/Y_0 = 1$. Moreover, the real interest rate is constant and equal to 2%. Finally, the primary deficit to GDP ratio is 3%, assumed to be constant over time. In order to keep stable the level of debt to GDP over time, the growth rate of the economy should be at least 4%.

Question 4

True or False?

Explain whether the following statements are true or false. Motivate your answer in a brief but rigorous way, by making explicit reference to the relevant theory. Lack of proper explanations will result in zero points.

The one-year interest rate in the US is 2%, the domestic interest rate in the euro area is 1% and the nominal exchange rate between the US dollar and the euro is 1.1 (the number of dollars to purchase a euro). Assuming the financial markets are perfectly operating, i.e. the interest parity condition holds, this means that markets expect an appreciation of the euro in the future.